

Embeddings and Semantic Search

An embedding represents text as a list of numerical values. An embedding model maps passages into a vector space where texts with related meanings tend to have similar representations. The numbers are learned features rather than explicit labels for individual concepts.

Semantic search compares these representations to find meaningfully related material. For example, a search for car repair may match a passage about automobile maintenance even when the exact words differ. This makes semantic search useful when users and authors express the same idea using different vocabulary.

A vector store keeps document embeddings together with identifiers and metadata. It supports similarity search and helps applications locate the corresponding source text. Metadata such as document name, topic, and access category can also filter results before they are returned to the application.

Indexing and Top-K Retrieval

Indexing prepares stored vectors for efficient similarity search. During ingestion, the application computes an embedding for each document passage and adds it to the vector store with its identifier and metadata. A search index can reduce the work required to compare a query against a large collection.

At search time, the application computes a query embedding using a compatible embedding model. Top-K retrieval returns the K highest ranked candidates under the configured similarity measure. A larger K supplies more candidates but may include less relevant material. Ranking alone does not guarantee that a candidate answers the question.

Document embeddings should be computed ahead of time and reused across queries. Recomputing unchanged passages wastes processing time and money. Changed passages require updated embeddings; changing embedding models may require rebuilding the collection's vectors.